Week of 28 September 2026

Idea path. One nominated paper. Grant-agnostic: no call names, no funding notes.

Chosen paper

Stratospheric ozone projections under sulfur-based stratospheric aerosol injection: Insights from the multi-model G6-1.5K-SAI experiment

Bednarz et al. (2026), Atmospheric Chemistry and Physics. Picked by Mrs. Lantern after the nomination passed.

https://doi.org/10.5194/acp-26-12751-2026

Takeaways

Bednarz et al. show model disagreement under G6-1.5K-SAI: UKESM dries the stratosphere where CESM and MIROC moisten it, and UKESM loses NH mid-latitude ozone where other models do not. Chemistry (WITH_HET / NO_HET) explains only part of the ozone change. The paper does not study solid particles.

Ideas

Idea 1

Why does UKESM dry the stratosphere under subtropical SO_2 SAI when CESM and MIROC moisten it? Bednarz lists four candidate processes beyond cold-point warming and asks for follow-up. This idea is that UKESM attribution.

Idea 2

Why does UKESM lose NH mid-latitude ozone under G6-1.5K-SAI while other models do not? Bednarz already shows chemistry is only part of it (WITH_HET / NO_HET). This idea digs into residual circulation versus mixing.

Idea 3

Can that same chemistry-versus-transport habit transfer to high-burden volcanic cases, and later to solid-particle SAI only if GLOMAP/UKCA is ready? Not claimed by Bednarz. The inferred tag stays.

Research directions

Direction A, from Idea 1

Take Bednarz's four candidate processes for UKESM's dry stratosphere under G6-1.5K-SAI and rank which ones matter in the model, using budgets and diagnostics already available.

Direction B, from Idea 2

Use the existing WITH_HET / NO_HET UKESM pair to split how much of the NH mid-latitude ozone drop is chemistry versus transport, and whether residual circulation or mixing does more of the transport work.

Direction C, from Idea 3

Reuse that chemistry-versus-transport mindset on high-burden volcanic cases first. Solid-particle SAI only after a written GLOMAP/UKCA readiness check.

Fences that hold: do not drop Idea 3’s inferred tag; do not claim Bednarz studied solid particles.

Mrs. Vanguard